
Formal-AVS: A Lean 4 Benchmark for Anytime-Valid Confidence-Sequence Theorem Proving

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Abstract

Anytime-valid confidence sequences (CS) give confidence intervals valid at every stopping time, removing the optional-stopping problem in A/B testing and clinical trial interim analyses. A Lean 4 proof pinned against a fixed Mathlib commit removes the class of implementation bugs that invalidate the nominal type-I guarantee. We release **FORMAL-AVS**: 60 Lean 4 theorems on anytime-valid CS across four families (Howard-Ramdas, betting, Whitehouse vector, asymptotic CLT), grouped into four pre-evaluation tiers T0–T3 and two empirical failure categories T4–T5. The companion library **formal-avs-lean** supplies the rate functions Mathlib does not yet provide.

We evaluate four closed-source generalists, two open-weights provers (DSPv2-7B, Goedel-V2-8B), and Harmonic Aristotle across three capability levels on a 48-target slate. Single-shot generalist closure peaks at 22/48 (Gemini, hint-list prompt). Under agentic mode (`cycle_prove r=4`), Gemini reaches 19/48 and closes the first non-Aristotle T3 cross-function target. Goedel-V2, trained with verifier-guided self-correction, rises from 5/48 single-shot to 13/48 agentic, surpassing DSPv2’s 9/48. The hint-list advantage narrows under agentic mode: Gemini’s gain vanishes ($19 = 19$), suggesting per-target compiler errors substitute for static prompt engineering. Aristotle closes 53/60 axiom-clean and refutes three false-as-stated targets. We release a 14-session Aristotle history archive and a 3-column T3 hint ablation alongside the benchmark.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

A/B tests on tech platforms run in the thousands per day. The classical fixed-horizon t -test assumes the sample size is chosen before looking at the data. In practice analysts peek. They stop experiments that look significant or extend those that look close, which inflates the false-positive rate from 5% to often 30% or higher. The same *optional stopping* problem arises with higher stakes in clinical trials. Interim analyses are essential there, but classical group-sequential designs require the number and timing of looks to be pre-specified. Anytime-valid confidence sequences (Howard et al., 2021; Waudby-Smith & Ramdas, 2024; Whitehouse et al., 2025) supply confidence intervals valid at every stopping time simultaneously, removing this constraint. The non-asymptotic flavour (Howard-Ramdas, betting, Whitehouse vector) gives hard guarantees at every t . The asymptotic flavour (time-uniform central limit theorem, CLT) gives the looser large-sample behaviour that governs deployment-scale runtime characteristics. The benchmark covers both regimes, since deployed systems span both.

The theory is clean. The code that computes the confidence band is not. A missing absolute value, an off-by-one on a log, a finite-precision coarsening of the cumulant bound, or a dropped symmetry hypothesis can each invalidate the nominal type-I guarantee while the software still appears to run. A Lean 4 proof pinned against a fixed Mathlib commit removes that class of error. Writing these proofs by hand is labour-intensive: each theorem takes a trained statistician-programmer hours to days. Closing them with language-model drafters would reduce the manual proof effort required for production deployment of anytime-valid methods. This paper benchmarks current drafters on a narrow, audited slice of that workload.

Existing Lean-proof benchmarks (MiniF2F (Zheng et al., 2022), ProofNet (Azerbayev et al., 2023), Verina (Ye et al., 2025)) measure the ability of a language-model drafter to close a proof against a frozen statement library. Given a frontier Lean library with a specific conceptual structure, which drafter capabili-

ties are required to close theorems about that structure, and can those capabilities be decomposed into prompt-level interventions? To answer this we release FORMAL-AVS, a benchmark of 60 Lean 4 theorems on anytime-valid confidence sequences (CS) (Howard et al., 2021; Waudby-Smith & Ramdas, 2024).

Main finding. The T3 tier (cross-family inequalities between rate functions) is a *reasoning gap*, not a knowledge gap. Supplying the log-domain substitution and the Mathlib lemma chain closes T3 to 3–5/6 per drafter; no drafter finds the chain autonomously (0–1/6 at baseline). This decomposition is the benchmark’s central result: the capability that separates Aristotle (6/6) from the best generalist (1/6) is not access to the right lemmas but the ability to discover the right rewrite without being told. The targets cover sign and monotonicity of family-specific rate functions, cross-function inequalities between the HR-family (Howard et al., 2021), betting family (Waudby-Smith & Ramdas, 2024), Whitehouse vector (Whitehouse et al., 2025), and asymptotic CLT families, and admissibility theorems for the four canonical stopping rules. A *target* in the rest of the paper means one of these 60 Lean theorem statements that a solver is asked to close. The companion library is released under Apache-2.0 at the snapshot URL listed in the supplementary archive (REPRODUCE.md, item LIB-SNAPSHOT). Quality assurance for the library is described in §3.2. It includes the axiom-clean baseline $\mathcal{A}_{\text{core}} := \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$ enforced on every closed proof, manual review of all ~ 50 dependency lemmas, and Aristotle-surfaced refutations of three statements.

The benchmark is paired with an Aristotle prover history archive. Fourteen sessions of Harmonic Aristotle (Achim et al., 2025) applied to targets drawn from the same library are released alongside the benchmark, with per-session verdicts (closed / partial / refuted), closed Lean proofs, counterexample witnesses, and `#print axioms` audits. Prior Lean benchmarks release target statements or solver traces, not both.

On a 48-target evaluated slate, four closed-source generalists, two specialised open-weights provers (DSPv2-7B (Ren et al., 2025), Goedel-Prover-V2-8B (Goedel-LM Authors, 2025)), and Harmonic Aristotle (Achim et al., 2025) are evaluated across three capability levels. Under single-shot $N=5$, generalist closure saturates at 22/48 (Gemini hint-list). T0 Mathlib-canon targets return 0/28 across four generalists. T3 cross-function targets return 2/24. Under agentic mode (`cycle_prove r=4` (Freer & Math Inc., 2025)), Gemini reaches 19/48 and closes the first non-Aristotle T3 target. The prompt-sensitivity gap narrows: Gemini’s

hint-list advantage vanishes under agentic mode, suggesting the refinement loop partially substitutes for prompt engineering (§4.3). Aristotle, under an unbounded compute budget, closes 53/60 axiom-clean and refutes three false-as-stated targets (§4.4). We also report a 3-column T3 hint study (§3.5) and a prompt-sensitivity matrix across all three capability levels (Table 2).

FORMAL-AVS has four single-shape tiers (T0-T3) plus two empirical failure categories. T4 contains four targets blocked by missing Mathlib APIs. T5 contains one unresolved finite-precision witness target. Tier shape: T0 (7), T1 (34), T2 (8), T3 (6), T4 (4), T5 (1). Each theorem is annotated with a JSON record containing statement, tier, required imports, per-drafter closure rates at $N = 5$, and a solver-status summary (§3.3). Appendix ?? (Listings ?? and ??) shows the schema and a representative T3 target. T4 and T5 assignments depend on observed solver behaviour and are flagged as such throughout the text.

This paper releases three artifacts. (i) `formal-avs-lean`, an Apache-2.0 Lean 4 library authored as a prerequisite for this benchmark ($\sim 3k$ lines, 8 modules, ~ 50 axiom-clean dependency lemmas) that instantiates the four canonical anytime-valid CS families and the rate functions required to even state the cross-function targets. (ii) A 60-theorem benchmark with T0-T3 single-shape tiers and T4-T5 empirical failure categories, plus per-drafter closure records and a 3-column T3 hint study. (iii) A 14-session Aristotle history archive against the same library, with closed/partial/refuted verdicts, closed Lean proofs, counterexample witnesses, and axiom audits. The refutation track measures a capability axis orthogonal to `pass@N` closure: whether a solver can identify false-as-stated theorems (§4.4).

2. Background and related work

Anytime-valid confidence sequences. A confidence sequence $(C_t)_{t \geq 1}$ at level $1 - \alpha$ for a parameter μ is a sequence of random intervals satisfying $\mathbb{P}(\forall t \geq 1 : \mu \in C_t) \geq 1 - \alpha$. The four canonical constructions for bounded-mean or sub-Gaussian settings each specify an exponential supermartingale $M_t = \exp(\lambda S_t - \psi_F(\lambda)t)$ with a family-specific cumulant-bound function ψ_F . Ville’s inequality applied to M_t produces a $1 - \alpha$ CS with half-width $\eta_F(\alpha, t) = (\psi_F^{-1} \circ \text{lin})(\alpha, t)$. The benchmark targets theorems about these four η_F (Howard et al., 2021; Waudby-Smith & Ramdas, 2024; Whitehouse et al., 2025), their monotonicity, sign, and cross-family inequalities.

Lean proof benchmarks. MiniF2F (Zheng et al., 2022) covers competition arithmetic. ProofNet (Azerbayev et al., 2023) covers undergraduate analysis, algebra, and topology. Verina (Ye et al., 2025) targets verifiable code generation with 189 Lean tasks. FORMAL-MATH (FormalMATH Authors, 2025) contains 5,560 Lean 4 theorems from olympiad to undergraduate level. LeanDojo (Yang et al., 2023) extracts training and evaluation theorems from Mathlib. REAL-Prover (REAL-Prover Authors, 2024) and LeanAgent (LeanAgent Authors, 2024) mine Mathlib-adjacent undergraduate content. FORMAL-AVS differs in target domain: a Lean benchmark focused on a research subfield not currently represented in Mathlib, with axiom-audited dependency proofs and a cross-family inequality tier.

Harmonic Aristotle and prover-history archives.

The companion archive uses Harmonic’s Aristotle prover (Achim et al., 2025), a closed-source proprietary Lean 4 proof-search engine with axiom-audit and counterexample-extraction hooks, and distinguishes genuine closures from drafter hallucinations via machine-checked Lean kernel certificates. Aristotle is not an open-weights model. We accessed it through the public API at `aristotle.harmonic.fun` at no cost during benchmark construction. Closure verdicts and counterexample witnesses are reproducible from the released session IDs, but re-running an Aristotle session against the released library requires a Harmonic API key. The closest prior artefact is LeanDojo’s tactic-trace dataset. Our companion artefact differs in three ways: (i) Aristotle sessions rather than Mathlib commits, (ii) refutation traces with counterexample witnesses, and (iii) per-session axiom-audits (Lean `#print axioms` output). The third property is important for evaluating prover trustworthiness: drafter-produced Lean proofs may close the goal but widen the axiom set (Yang et al., 2023).

Hint-ablation methodology. Prompt-hint ablation for Lean drafters has been studied in the context of tactic-premise selection (Polu & Sutskever, 2020; Han et al., 2022; Jiang et al., 2023) and proof-sketch scaffolding. Most prior studies treat hints as a single “context” axis and measure gross closure-rate gain. FORMAL-AVS decomposes the hint into two orthogonal components, a log-domain substitution hint (*lateral re-encoding*) and a Mathlib-lemma-chain hint (*tactic vocabulary*), and measures their effects independently across four drafter classes plus one specialised 7B prover. The 3-column ablation (§4) shows neither component alone is sufficient; closure rates increase only when both are supplied to mid-tier generalists.

Agent-style evaluation of code and proof drafters. Single-shot benchmarks understate drafter capability: the same models close many more issues when granted file navigation, multi-step tool calls, and execution feedback, as shown by SWE-bench (Jimenez et al., 2024; Chowdhury et al., 2024). Agentic frameworks for theorem proving and code/database editing are emerging in parallel (Liu et al., 2024; Wang et al., 2024). FORMAL-AVS supports both completion- and agent-style evaluation. A target’s JSON record exposes the tier shape, required imports, and the $\mathcal{A}_{\text{core}}$ axiom audit, which are the inputs an agentic evaluation loop consumes when delegating to a Lean kernel. The headline sweep is the completion-style baseline against which any agentic-mode gain can be measured. We do not run an agent loop in this submission, but the schema (§??) and `sweep.py` driver are designed so the same target slate can be re-run under one without re-curation.

Anytime-valid CS formalisations. No prior public Lean 4 library formalises the HR-family, betting family, Whitehouse vector, or asymptotic CLT confidence-sequence families. The `mathlib Probability.Martingale` module (The Mathlib Community, 2026) supplies Doob decomposition and Azuma’s inequality but does not instantiate the four CS families or their sharp-constant comparison inequalities. The `formal-avs-lean` library (released with this benchmark) instantiates all four families, supplies Ville’s inequality and the quantization-transport lemma, and exposes 29 theorems that require library-specific definitions.

3. Benchmark Construction

We define a *target* as a single Lean 4 theorem statement that a solver is asked to close. FORMAL-AVS contains 60 targets (§3.3). Each is annotated with its tier, required imports, family, and per-drafter closure record. Appendix ?? (Listings ?? and ??) shows one target as Lean source and as a JSONL row. The field-level schema is in Appendix ??.

3.1. The companion library: `formal-avs-lean`

We construct `formal-avs-lean` (Anonymous, 2026), an Apache-2.0 Lean 4 library released as a prerequisite artefact for this benchmark. It instantiates the four canonical anytime-valid confidence-sequence constructions, supplies per-family rate functions η_F , and proves the dependency lemmas required to even *state* the cross-family inequalities used in T3. The asymptotic CLT family is included for comparability with deployed

systems that mix non-asymptotic and asymptotic CS constructions. The library is organised into eight modules (`Basic`, `HowardRamdas`, `WaudbySmithBetting`, `WhitehouseVector`, `AsymptoticCLT`, `Quantization`, `PhiTransform`, `MatchingConstants`). Target theorems are extracted from each.

FORMAL-AVS contains 60 targets by design: target curation is bounded by the ~ 50 axiom-clean dependency lemmas a library author can hand-prove, which allows manual audit of axiom use. Translation pipelines sourcing olympiad or textbook content cannot reach the cross-family tier because Mathlib does not instantiate the rate functions.

Why formal-avs-lean was necessary. Mathlib’s `Probability.Martingale` supplies Doob decomposition and Azuma’s inequality but does not instantiate the four CS families, the per-family rate functions, or Ville’s inequality for the betting-process construction. Without those, neither generalist drafters nor Aristotle can attempt the T3 cross-function statements: the goal type $\eta_{\text{HR}}(b) \leq \eta_{\text{vector}}(b)$ is not expressible. The library closes that gap in $\sim 3\text{k}$ lines across 8 modules and ~ 50 dependency lemmas, with the QA guarantees described in §3.2.

3.2. Quality assurance and axiom audit

Because the benchmark grades drafters against the library, the library itself must be trustworthy: a hidden `sorry` or a silently-widened axiom set in any dependency lemma would invalidate every closure built on top. Quality assurance has four components. (i) **Axiom-clean baseline.** Every reference proof, every closed drafter attempt, and every closed Aristotle session terminates inside the canonical Mathlib core $\mathcal{A}_{\text{core}} = \{\text{propxt}, \text{Classical.choice}, \text{Quot.sound}\}$. The drafter-sweep closure criterion (§3.3) rejects any candidate whose `#print axioms dump` extends beyond $\mathcal{A}_{\text{core}}$, and per-target dumps union to exactly $\mathcal{A}_{\text{core}}$ (Appendix ??). This rules out the axiom `my_lemma : ...` shortcut. (ii) **Hand-authored dependency lemmas.** The ~ 50 dependency lemmas (Ville’s inequality, quantization-transport, Φ -substitution, sharp constants) were written by the library authors before any solver run, and each was manually cross-checked against the published constructions of Howard et al. (Howard et al., 2021), Waudby-Smith and Ramdas (Waudby-Smith & Ramdas, 2024), and Whitehouse et al. (Whitehouse et al., 2025). (iii) **Refutation discovery.** Three benchmark candidates authored as legitimate targets were refuted by Aristotle on the first attempt, and both original and corrected forms ship in the JSONL (§4.4). (iv) **Toolchain pinning** (§??): library, JSONL, and

verification driver are pinned to a fixed Lean toolchain and Mathlib commit, and `bash reproduce.sh` re-runs the full audit on cached outputs.

The dependency lemmas are sorry-free. Target theorems contain `sorry` by design: these are the statements solvers attempt to close, and the 53/60 closed proofs are in the Aristotle history and drafter output files.

3.3. Tiers and failure categories

Each target carries two kinds of label. The four *tiers* T0-T3 are assigned by theorem shape before any drafter run. The two *empirical failure categories* T4 and T5 are assigned after evaluation, because their definitions reference observed solver behaviour. We separate the two kinds of label so that the tier signal is independent of solver outcome.

Pre-evaluation tiers (T0-T3). **T0** (Mathlib canon, 7 theorems): statement and proof expressible entirely within Mathlib (sanity-check floor). **T1** (single-function friendly, 34): sign / monotonicity / positivity of a single η_F closeable by `positivity`, `nlinarith`, or a short `simp-chain`. **T2** (single-function challenging, 8): single-function targets whose closure requires a substitution or a Mathlib lemma not named in the goal. **T3** (cross-function inequality, 6): inequalities or sharp-constant identities between two rate functions (e.g. $\eta_{\text{vector}} = \sqrt{2} \eta_{\text{HR}}$), admitting a log-domain Φ -substitution. T0-T3 assignment depends only on theorem signature and is fixed before any drafter sweep.

Empirical failure categories (T4, T5). **T4** (library-gap, 4 theorems): targets where no evaluated solver returned a closure and the missing piece is a Mathlib API not yet upstream (sub-Gaussian MGF bound for martingale sums, Ville’s inequality for supermartingales, vector concentration, time-uniform CLT). A Mathlib pull request adding the required API would move these targets out of T4. **T5** (unresolved, 1 theorem): the sole T5 target, `equivalence_break_at_finite_precision`, has zero close-rate across every evaluated solver. It is a witness-construction problem on the continuous-equivalence break between Howard-Ramdas and Ville thresholds at finite precision; no missing Mathlib API explains the failure. T4 and T5 are empirical categories from this evaluation, not predictions for future solvers.

3.4. Drafter sweep protocol

Each drafter is evaluated at $N = 5$ attempts per target under a mixed-temperature schedule: one greedy attempt (temperature $T = 0$) followed by four inde-

pendent samples at $T = 0.7$. The attempt budget is 8k tokens per sample. We define $\text{pass}@N$ at the target level. A target is *closed* for a drafter if at least one of the N attempts produces a Lean 4 proof that (a) type-checks on the specified toolchain, (b) closes the goal with no residual `sorry`, and (c) does not widen the axiom set beyond `{propext, Classical.choice, Quot.sound}`. $\text{Pass}@N$ is defined at the target level: a target is closed if any of the N attempts succeeds. We do not aggregate over individual attempts as if they were independent samples.

Target-selection protocol. We extracted target theorems from the eight `formal-avs-lean` modules. The selection criteria, applied in order, were the following. (i) The theorem is provable in Lean 4 against Mathlib at the pinned commit, modulo the dependency lemmas already proved inside `formal-avs-lean`. (ii) The statement does not import a non-Apache-2.0 third-party library. (iii) The statement is at most 400 characters and does not require a custom tactic block to state. We did not pre-screen for drafter difficulty. Of the ~ 120 candidate theorems in the library, 60 met the criteria and form the released benchmark. The remaining candidates were excluded for reasons recorded per-target in the supplementary `selection_log.csv`.

Evaluated slate. The headline drafter sweeps were run on a 48-target subset. The 48-slate is the canonical evaluation set in this submission. The full 60-target table adds 4 T4 entries, 1 T5 entry, and 7 T1 entries that we report separately (see Table 1, footnote on imputed cells). T4 and T5 entries are zero by construction in the 60-target table because they were assigned to those categories on the basis of zero closures across evaluated solvers (§3.3). We report headline rates on the 48-slate and a status table on 60 targets, rather than fold the two together.

Drafter set. Four closed-source drafters are scored in the headline sweep. Claude Sonnet 4.6, Kimi K2.5, and Gemini 3 Pro under a shared hint-list prompt. Mistral Large 3 under a neutral prompt (§4.2 reports the prompt-sensitivity result). Mistral collapses to 0/240 on the same hint-list prompt because of greedy hint-list compliance, so the headline row uses the prompt that produces a non-degenerate row. Cross-drafter aggregates over Mistral and the other three should be read with this prompt asymmetry in mind. A supplementary closed-source row, Claude Opus 4.6, is reported on the same 48-slate at $N = 5$ under the hint-list prompt. Two specialised open-weights drafters complete the headline sweep: DeepSeek-Prover-V2-7B (Ren et al., 2025) (huggingface.co/deepseek-ai/DeepSeek-

Prover-V2-7B) and Goedel-Prover-V2-8B (Goedel-LM Authors, 2025) (huggingface.co/Goedel-LM/Goedel-Prover-V2-8B). Both are scored on the 48-slate at $N = 5$ under the same neutral specialised-prover prompt template. Community quantization choices and reproduction details are in Appendix ?? . DSPv2 attempts are released as cached candidates in the supplementary archive. The Goedel-V2 row is reproducible end-to-end from the released weights and evaluation driver.

3.5. Hint ablation protocol

Three prompt columns per T3 target are evaluated, with names chosen so the content is recoverable from the label. **Base:** goal statement only (no-hint baseline). **Sub:** BASE plus a one-line Φ -substitution hint $x \mapsto \exp(\lambda^*x - (\lambda^*)^2\sigma^2/2)$, with family-specific λ^* (isolates lemma search given the substitution). **Sub+Chain:** SUB plus an explicit Mathlib lemma chain (`Real.exp_le_one_iff`, `Real.log_nonneg`, `Real.sqrt_le_sqrt`, ...) with usage-direction annotations (isolates lemma application given both). A boxed-table form is Table ?? (Appendix ??). The five-stage evaluation pipeline is expanded as Table ?? in Appendix ??.

The SUB+CHAIN lemma list was reverse-engineered from an Aristotle T3 closure (§3.3, footnote on hint-chain provenance). An alternative library-reference chain (§??) is released as a sensitivity check, overlapping on 5/7 lemmas. We read SUB+CHAIN as a proof-plan-condition upper bound on drafter T3 closure given the lemma chain, not as an independent tactic-vocabulary measurement.

3.6. Aristotle history collection

Fourteen Harmonic Aristotle (Achim et al., 2025) sessions were collected between 2026-03 and 2026-04 against the same `formal-avs-lean` library. Each session submits 1 to 5 targets. We record per-target verdict (*closed*, *partial*, or *refuted*), the closed Lean proof when applicable, the counterexample when applicable, the axiom-set dump of the closed Lean kernel, and the fingerprint hash of the submitted Lean tarball. All 14 sessions are released as `aristotle_history.jsonl` alongside the benchmark. Reproducing a session against the released library requires a Harmonic API key (we accessed Aristotle through the public dashboard at `aristotle.harmonic.fun`).

Refutation handling. Three benchmark targets were authored as legitimate library statements but

Aristotle returned counterexamples on first attempt, identifying that the as-stated forms were mathematically false (§4.4). We retain all three in the benchmark JSONL. The released schema includes both the corrected theorem (used in headline drafter sweeps and the Aristotle 53/60 closure count) and the original refuted form (used to support the statement-correctness classification axis). Refutations were not seeded by the authors. They are a discovered property of the library that the benchmark surfaces and preserves.

Aristotle compute budget. Each Aristotle session is a single API call submitted via `aristotle submit` against the released library. We did not impose a per-session token or rollout cap. Sessions return on average within 30 to 50 minutes of wall time. Drafters are scored at $N = 5$ attempts of 8k tokens each. The Aristotle vs. drafter comparison is therefore between a proof-search system with unbounded internal compute and a generalist drafter at a fixed token budget. The 53/60 vs. 2/24 T3 gap conflates training data, training scale, search compute, and prompt-conditioned tactic vocabulary, and is reported as such.

Reproducibility. The supplementary archive includes the `formal-avs-lean` library at the cited Lean toolchain and Mathlib pin (toolchain reference and Mathlib commit are listed in the supplementary `REPRODUCE.md`), the benchmark JSONL with both corrected and original (refuted) statement forms, the prompt files, the drafter sweep driver, the Wilson-CI computation script, and the Lean kernel verification script that recomputes the `{propext, Classical.choice, Quot.sound}` axiom audit. A single command (`bash reproduce.sh`) runs the verification pipeline on cached drafter outputs to regenerate every numeric cell in the headline tables. Drafter output files for all evaluated cells are released in the archive.

4. Results

4.1. Per-tier results on the evaluated 48-slate

Table 1 reports per-drafter closure rates at $N = 5$ on the evaluated 48-slate (T0-T3 shapes), plus a 60-target status table. T4 and T5 entries are zero in the 60-target table by construction (§3.3). Aristotle’s closure rate is in §4.1. Most drafter closures fall on T1; the other five tiers show low closure.

T0 (0/28 across 4 generalists): generalists fail on Mathlib-canon lemmas despite these appearing in training corpora. The failure mode is a tactic-prior mismatch: drafters attempt multi-step rewrites

Table 1. Per-drafter closure rates on the 48-slate at $N=5$ (one greedy + four $T=0.7$ samples). Overall column sums the 60-target status table (12 extras 0-imputed; T4/T5 zero by construction). Wilson CIs in Appendix ?? . See Appendix ?? for $N=10$ results.

Drafter	T0	T1	T2	T3	T4	T5
Claude Sonnet 4.6	0/7	18/34	0/8	0/6	0/4	0/1
Kimi K2.5	0/7	17/34	0/8	1/6	0/4	0/1
Gemini 3 Pro	0/7	22/34	0/8	0/6	0/4	0/1
Mistral Large 3*	0/7	1/34	0/8	1/6	0/4	0/1
Claude Opus 4.6 ^{††}	0/7	13/34	0/8	0/6	0/4	0/1
DSPv2-7B GPTQ-int8 [§]	3/7	8/34	0/8	1/6	0/4	0/1
Goedel-Prover V2-Q6_K [¶]	2/7	3/34	0/8	0/6	0/4	0/1

*Mistral on neutral prompt (0/240 on hint-list). ^{††}Opus on hint-list only. [§]DSPv2 on neutral prompt. [¶]Goedel-V2 GGUF Q6_K on neutral prompt. [†]pass@10 adds 3 new closures.

where a single `simp` or `exact?` suffices, because the `formal-avs-lean` namespace context shifts the tactic distribution away from the Mathlib patterns the models memorised (§??). **T2** (0/8): requires a substitution or unnamed Mathlib lemma. **T3** (2/24): drafter closure 0–1/6. Aristotle closes 6/6. The gap is a system difference, not per-token (§4.3). **T4** (0/4): blocked by missing Mathlib APIs (Ville, sub-Gaussian MGF, vector concentration, time-uniform CLT). **T5** (0/1): Aristotle refuted the as-stated form (§4.4). Corrected form unresolved.

Capability ladder: single-shot → agentic → Aristotle. Three capability levels are evaluated on the same slate.

Hint ablation (T3). A 3-column T3 hint study (BASE, SUB, SUB+CHAIN) is reported in Appendix ?? . Generalist closure on T3 rises from 0–1/6 at BASE to 3–5/6 at SUB+CHAIN. The SUB+CHAIN chain was reverse-engineered from an Aristotle closure, so we read it as an upper bound (sensitivity check in Appendix ??).

4.2. Prompt-sensitivity is drafter-specific and bidirectional

Table 2 reports per-drafter closure rate under both the shared hint-list prompt (which enumerates nine allowed Mathlib tactics as suggestions and asks for minimal proofs) and a neutral prompt that omits the enumeration (“Close this theorem with a clean Lean 4 proof. Use any Mathlib tactic.”). Both evaluated on the same 48-target drafter-testable slate at $N = 5$ (one greedy $T = 0$ + four $T = 0.7$).

Table 2. Capability ladder: prompt-sensitivity across single-shot, agentic, and Aristotle on the 48-slate (pass@5). Single-shot uses $N=5$ (1 greedy + 4 stochastic). Agentic uses `cycle_prove` $r=4$ (draft $\rightarrow$ lake build $\rightarrow$ refine, default from OpenGauss (Freer & Math Inc., 2025)). Hint-list enumerates nine Mathlib tactics. Neutral omits the enumeration. Aristotle (Achim et al., 2025) operates under an unbounded compute budget (no prompt distinction). The hint-list advantage shrinks under agentic mode: Sonnet’s $6\times$ single-shot lift ($3 \rightarrow 18$) reduces to 15% ($13 \rightarrow 15$). Gemini’s advantage vanishes ($19 = 19$): per-target lake errors substitute for the static tactic enumeration.

Drafter	Hint-list	Neutral
<i>Single-shot ($N=5$)</i>		
Gemini 3 Pro	22/48	13/48
Claude Sonnet 4.6	18/48	3/48
Kimi K2.5	18/48	13/48
DSPv2-7B	12/48	9/48
Goedel-V2-Q6_K	5/48	3/48
Mistral Large 3	0/48	2/48
<i>Agentic (<code>cycle_prove</code> $r=4$)</i>		
Gemini 3 Pro	19/48	19/48
Claude Sonnet 4.6	15/48	13/48
Goedel-V2-Q6_K	10/48	13/48
Aristotle (unbounded)	53/60	

The hint-list helps three of four generalists and reduces Mistral ($2/48 \rightarrow 0/48$). Sonnet’s sensitivity is largest ($6\times$). Extended per-drafter analysis in Appendix ??.¹

4.3. Capability ladder: single-shot, agentic, and Aristotle

Closure rate increases monotonically across three capability levels on the same slate: *single-shot* (best-of- N , no error feedback), *agentic* (draft $\rightarrow$ lake build $\rightarrow$ feed errors back $\rightarrow$ refine, $r=4$ rounds), and *Aristotle* (unbounded internal refinement). Harmonic Aristotle (Achim et al., 2025) closes 53/60 axiom-clean (7/7 T0, 33/34 T1, 7/8 T2, 6/6 T3) plus 3 refuted (§4.4). Companion-archive closures are in §???. Under `cycle_prove` $r=4$:

- **Gemini 3 Pro.** Single-shot neutral 13/48 $\rightarrow$ agentic 19/48 (46% lift). Gemini agentic closes `ranking_four_way` (T3) on round 2: the first non-Aristotle closure on a cross-function target.
- **Sonnet 4.6.** Neutral 3/48 $\rightarrow$ agentic 15/48 ($5\times$ lift from the refinement loop alone).
- **Goedel-V2-Q6_K.** Single-shot 5/48 $\rightarrow$ agentic 13/48 ($2.6\times$ lift). Goedel-V2 was trained with

¹Four extended analyses are in Appendix ??: Aristotle history, companion closures, cross-drafter tactic diversity, and the no-closure tail.

verifier-guided self-correction (Goedel-LM Authors, 2025): the model uses Lean compiler feedback to iteratively revise its proofs. Under `cycle_prove` this training activates and Goedel surpasses DSPv2’s single-shot 9/48, becoming the strongest local prover on our slate at \$0 marginal cost.

The prompt-sensitivity gap narrows under `cycle_prove`. Sonnet’s single-shot $6\times$ hint-list advantage ($3/48 \rightarrow 18/48$) reduces to 15% under agentic ($13 \rightarrow 15$). Gemini’s advantage disappears ($19 = 19$). Per-target lake errors provide stronger signal than a static tactic list, partially substituting for prompt engineering.

Across both drafters, 14 targets close under both, 5 are Gemini-only (including 2 T3 cross-function targets), 1 is Sonnet-only. Most closures land on rounds 1–2 (28/34). Rounds 3–4 contribute 6/34: the refinement loop’s value is primarily one error-correction cycle, not deep multi-round search. Generalist single-shot saturates below 22/48, agentic $r=4$ reaches 19/48, and Aristotle’s open-ended budget closes 53/60. The hint ablation (Table ??) shows the tactic vocabulary is latent in the generalist drafters; the gap is attributable to search depth, not missing knowledge.

4.4. Refutation findings

Beyond closure, Aristotle refuted three benchmark targets whose as-stated form is mathematically false, returning machine-checkable counterexamples on each: `equivalence_break_at_finite_precision` (T5, missing symmetry hypothesis), `slack_antitone_in_scale` (T2, missing bits-match hypothesis, counterexample $\text{bp}_1=(1,1)$, $\text{bp}_2=(2,1)$), and `div_le_div_of_nonneg_left` (T1, typo in direction: $a=1, b=2, c=3$ gives $1/2 > 1/3$). Each corrected form was subsequently proved.

Provenance of the three refutations. These statements were authored as legitimate library targets prior to any solver run. Aristotle subsequently flagged each as-stated form as false on its first attempt. The refutations were not seeded into the benchmark by the authors. The benchmark JSONL contains the corrected forms in the headline target column and the original false-as-stated forms in a parallel `refuted_form` field.

The mix of correct, borderline, and false-as-stated targets therefore supports a second evaluation axis: *statement-correctness classification* via refutation. Pass@ N solvers that do not attempt counterexample construction miss this axis entirely.

5. Discussion

Why T3 failures persist after the substitution hint. Every T3 target reduces to a single-function inequality that one of the four drafters could close autonomously once the right rewrite is in hand. Two operations are jointly required for closure: producing the log-domain Φ -substitution (or a problem-specific `sqrt` manipulation), and selecting the Mathlib lemma chain that discharges the re-encoded goal. The SUB column of Table ?? (0-2/6 per drafter) isolates the second operation. The SUB+CHAIN column (3-5/6) supplies both. Harmonic Aristotle closes 6/6 (§4.1) without either hint, using the Φ -substitution on some targets and a direct `sqrt.le_sqrt`-family route on others. Aristotle is under a different compute regime (§3.3). The 6/6 vs 0-2/6 gap should be read as a system difference, not as an isolated generalist tactic-vocabulary bound. The SUB+CHAIN result suggests supplying the lemma list narrows the generalist closure gap on T3, conditional on the Aristotle-derived chain (§3.3, hint-ablation paragraph).

Specialised provers and the Mathlib-outside gap. Goedel-Prover-V2 (Goedel-LM Authors, 2025), trained on Mathlib, surfaces this gap in its own reasoning trace: on `etabetting_nonneg` ($0 \leq \eta_{\text{betting}}(b)$), the model responds “*the Lean 4 code provided does not define etaBetting ... we cannot proceed with a meaningful proof.*” Under single-shot, DSPv2 (Ren et al., 2025) (9/48) outperforms Goedel-V2 (5/48) because its autocomplete format fits a 10 GB GPU. Under agentic mode, Goedel-V2’s self-correction training activates and it reaches 13/48, surpassing DSPv2 (detailed hardware-constraint analysis in Appendix ??).

Hint-response asymmetry. The T3 hint ablation (Appendix ??) shows generalist closure rising from 0-1/6 at BASE to 3-5/6 at SUB+CHAIN, while DSPv2 closes only one target across both hint columns. For mixed-drafter ensembles this suggests routing lemma-rich prompts to generalists and specialised provers to targets where their tactic prior matches the goal shape.

Limitations. Wilson 95% CIs, scope, and contamination notes are in Appendix ??.

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